

CARDINHAM BODMIN, United Kingdom

WMO: 038230

Lat: 50.5022N

Lon: 4.6669W

Elev: 653

StdP: 14.35

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.1	30.5	22.7	17.5	31.7	25.0	19.5	32.9	33.4	45.6	30.4	44.9	11.2	40	0.672

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	73.1	63.9	69.9	62.0	67.2	60.7	65.2	70.8	63.5	67.7	62.1	65.5	9.7	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	88.8	66.7	61.9	84.8	64.9	60.7	81.3	63.5	30.4	70.8	29.1	67.9	28.2	65.7	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
28.1	25.3	22.8	DB	25.6	77.9	3.6	2.9	23.0	80.0	20.9	81.7	18.9	83.4	16.3	85.5
			WB	24.6	68.7	3.4	2.2	22.2	70.3	20.2	71.6	18.4	72.8	15.9	74.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.2	42.0	42.0	44.2	47.2	52.0	56.3	59.7	59.5	56.7	52.3	46.9	43.2
	DBStd	7.81	5.11	4.87	4.55	4.55	4.59	4.15	3.85	3.20	3.39	4.02	4.48	5.29
	HDD50	1144	248	225	185	106	29	2	0	0	0	22	111	216
	HDD65	5417	712	644	646	535	404	263	173	176	250	393	544	677
	CDD50	1222	1	1	5	22	90	192	301	293	201	95	17	4
	CDD65	16	0	0	0	0	0	3	8	4	1	0	0	0
	CDH74	71	0	0	0	0	1	17	43	9	1	0	0	0
	CDH80	3	0	0	0	0	0	1	2	0	0	0	0	0
(14) Wind	WSAvg	11.4	13.3	12.9	12.6	11.5	11.0	10.0	9.5	9.6	10.2	11.6	11.9	13.0
(15) Precipitation	PrecAvg	47.5	5.3	4.1	2.9	2.9	2.8	2.8	3.3	3.6	3.3	5.2	5.5	5.3
	PrecMax	60.7	9.2	9.3	5.1	6.5	5.7	6.3	9.6	6.9	6.3	10.4	10.5	11.8
	PrecMin	37.8	0.8	0.9	0.9	0.7	0.3	0.4	0.4	0.7	0.6	1.9	2.6	1.6
	PrecStd	6.7	2.0	2.0	1.2	1.6	1.5	1.6	2.0	1.7	1.6	1.9	2.1	2.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.5	53.3	59.4	67.2	71.4	76.5	78.2	75.0	71.0	64.0	57.8	54.9
		MCWB	52.5	51.0	51.1	55.4	61.1	64.9	65.9	67.0	63.3	59.1	56.6	53.9
	2%	DB	52.0	51.6	54.8	61.0	66.7	70.8	74.3	70.2	67.2	60.9	56.0	53.8
		MCWB	51.2	49.9	49.1	53.2	58.5	61.8	64.6	63.8	61.3	57.8	54.8	53.0
	5%	DB	50.4	50.1	52.1	56.9	63.0	67.3	70.4	67.3	64.6	59.3	54.8	52.5
		MCWB	49.4	48.6	48.7	50.7	56.5	59.9	62.3	61.9	59.8	57.2	53.6	51.6
	10%	DB	49.2	48.8	50.3	54.0	59.7	64.1	66.8	65.0	62.6	58.1	53.6	51.0
		MCWB	48.1	47.4	48.0	49.2	54.7	58.2	60.8	60.6	58.6	56.3	52.3	49.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.9	52.3	52.9	57.5	62.9	66.0	67.4	68.0	64.9	60.9	57.3	54.5
		MCDB	53.1	52.9	55.7	63.8	68.3	75.4	75.8	73.0	68.2	62.4	57.7	54.6
	2%	WB	51.4	50.4	51.1	54.0	59.8	62.8	65.3	64.9	62.9	59.1	55.4	53.2
		MCDB	51.8	51.1	53.2	59.3	65.0	68.8	72.4	68.4	65.5	60.0	55.8	53.6
	5%	WB	49.7	49.1	49.8	51.9	57.5	60.9	63.3	63.2	61.2	57.9	54.0	51.9
		MCDB	50.2	49.8	51.3	55.4	61.9	65.9	68.7	65.8	63.3	58.8	54.5	52.4
	10%	WB	48.4	47.8	48.6	50.3	55.4	59.0	61.8	61.8	59.8	56.8	52.6	50.2
		MCDB	49.0	48.6	50.0	53.2	58.8	62.8	65.7	64.0	61.8	57.7	53.3	50.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.8	10.1	11.9	12.4	12.4	12.0	11.3	11.3	9.3	8.4	8.1
		MCDBR	8.3	8.8	12.1	15.8	16.6	17.5	18.2	14.2	13.7	9.8	8.1	8.1
		MCWBR	8.7	7.8	8.7	10.5	10.9	11.4	10.7	9.1	9.1	7.5	7.6	8.6
	5% WB	MCDBR	8.1	8.2	10.1	13.5	15.0	15.4	15.4	12.2	11.6	8.6	7.3	8.1
		MCWBR	8.7	8.0	8.3	9.5	10.4	10.7	9.9	8.7	8.8	7.6	7.6	8.6
(40) Clear-Sky Solar Irradiance	taub		0.326	0.339	0.365	0.374	0.381	0.369	0.372	0.358	0.354	0.347	0.325	0.317
	taud		2.373	2.402	2.341	2.339	2.354	2.412	2.409	2.467	2.468	2.460	2.422	2.337
	Ebn at Noon		221	252	265	275	278	282	279	278	266	246	221	203
	Edh at Noon		21	26	32	36	37	35	35	31	28	24	19	18
(44) All-Sky Solar Radiation	RadAvg		252	505	867	1393	1690	1866	1700	1451	1089	614	323	202
	RadStd		24	48	93	93	111	164	121	95	72	42	27	23

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page